Department of Information Technology: Gauhati University:

Curriculum for All Five Years of UG and PG Programme B.Sc. and M.Sc. in Software Development as per NEP 2020 structure: (As recommended by CCS held on 3rd May, 2024)

Name of the Programme: Integrated B.Sc. and M.Sc. in Software Development
Eligibility: 12 Grade Passed

Programme Specific Outcomes (PSOs):

PSO Number	PSO
PSO-1	To transform and empower graduates to meet global challenges through holistic education in terms of recent Teaching-Learning methodologies Specific to Software Development
PSO-2	To groom the graduates towards excellence through building communication skills, handling team and leadership challenges and negotiating career path ways specific to Software Development, and for the profession in general
PSO-3	To heighten the conscious of the graduates on socio-economic concern and to inculcate moral and ethical values to chisel them as better human being, being in the Software and It world
PSO-4	To train the student on the state-of-the-art tools and techniques and facilitate them to comprehend, analyze, design and create feasible solutions/innovative products in the field of Software Development for industries and real life problems
PSO-5	To pursue higher studies with good knowledge in core areas of Information Technology, through training on contemporary tools, techniques and technologies

B.Sc. in Software Development (3 Years):

Sem	Paper Code	Paper Name	Credit	Exit
1	SEC01XXX03	Skill Enhancement Course (SEC)	3	
	MDC01XXX03	Multi Disciplinary Course (MDC)	3	
	AEC01XXX02	Ability Enhancement Course (AEC)	2	
	VAC01XXX02	Value Added Course (VAC)	2	
	SWD0100104	Fundamentals of Computing	4	
	SWD0100204	Java Programming-I	4	
		Minor: Core-3	4	
		Semester Credit:	**22**	
	SEC02XXX03	Skill Enhancement Course (SEC)	3	
	MDC02XXX03	Multi Disciplinary Course (MDC)	3	
	AEC02XXX02	Ability Enhancement Course (AEC)	2	

2	VAC02XXX02	Value Added Course (VAC)	2	**1 Year Certificate in Software Development**
	SWD0200104	Programming for Problem Solving	4	
	SWD0200204	Java Programming-II	4	
		Minor: Core-3	4	
		Semester Credit:	**22**	
3	SEC03XXX03	Skill Enhancement Course (SEC)	3	**2 Years Diploma in Software Development**
	MDC03XXX03	Multi Disciplinary Course (MDC)	3	
	XXX	Internship	4	
	SWD0300104	Data Structure and Algorithm	4	
	SWD0300204	Database Management System	4	
		Minor: Core-3	4	
		Semester Credit:	**22**	
4	VAC04XXX02	Value Added Course (VAC)	2	
	SWD0400104	Software Engineering	4	
	SWD0400204	Data Communication and Computer Network	4	
	SWD0400304	Web Technology	4	
	SWD0400406	Software Product Development Project (GUI based)	6	
		Semester Credit:	**20**	
5	AEC05XXX02	Ability Enhancement Course (AEC)	2	**3 Years B.Sc. in Software Development**
	SWD0500104	Operating System (3-1-0)	4	
	SWD0500204	Advanced Web Technology (3-1-0)	4	
	SWD0500304	System Administration (3-0-1)	4	
	SWD0500404	UX Design (3-0-1)	4	
	SWD0500504	Full Stack Software Development Lab	4	
		Semester Credit:	**20**	
6	AEC06XXX02	Ability Enhancement Course (AEC)	2	
	SWD0600104	Python Programming	4	
	SWD0600204	Mobile Application Development	4	
	SWD0600304	Cybersecurity	4	
	SWD0600406	Full Stack Product Development Project	6	
		Semester Credit:	**20**	
		Total Credits:	**126**	

B.Sc. (Honours) in Software Development (4 Years):

Sem	Paper Code	Paper Name	Credit	Exit
7	SWD0700104	Research Methodology	4	
	SWD0700204	Software Testing and Quality Assurance	4	
	SWD0700304	Artificial Intelligence and Machine Learning	4	
	SWDXXXXX04	Elective-1	4	
	SWD0700404	Honours Project	4	

		Semester Credit:	**20**	
8	SWD0800104	Software Project Management	4	**B.Sc.. (Honours) in Software Development**
	SWD0800204	Cloud Services and Management	4	
	SWDXXXXX04	Elective-2	4	
	SWDXXXXX04	Elective-3	4	
	SWD0800304	Honours Project with Seminar	4	
		Semester Credit:	**20**	
		Total Credits:	**166**	

B.Sc. (Honours with Research) in Software Development (4 Years):

Sem	Paper Code	Paper Name	Credit	Exit
7	SWD0700104	Research Methodology	4	**B.Sc. (Honours with Research) in Software Development**
	SWD0700204	Software Testing and Quality Assurance	4	
	SWD0700304	AI and ML	4	
	SWDXXXXX04	Elective-1	4	
	SWD0700404	Honours Project	4	
		Semester Credit:	**20**	
8	SWD0800404	Seminar	4	
	SWD0800516	Dissertation	16	
		Semester Credit:	**20**	
		Total Credits:	**166**	

1 Year PG: M.Sc. in Software Development (5th Year):

M.Sc. (Research) in Software Development

Sem	Paper Code	Paper Name	Credit	Exit
9	SWD0900120	Dissertation Phase 1	20	**M.Sc. (Research) in Software Development**
10	SWD1000120	Dissertation Phase 2	20	
		Total Credits:	**40**	

M.Sc. (Course Work) in Software Development

Sem	Paper Code	Paper Name	Credit	Exit
9	SWD0900204	Deep Learning	4	**M.Sc. (Course Work) in Software Development**
	SWD0900304	Enterprise Resource Planning	4	
	SWD0900404	Internet of Things: Software Development for Embedded Systems	4	
	SWDXXXXX04	Elective-1	4	
	SWD0900504	Masters Project Phase-1	4	
		Semester Credit:	**20**	
10	SWD1000204	Data Mining and Data Visualization	4	
	SWDXXXXX04	Elective-2	4	
	SWDXXXXX04	Elective-3	4	
	SWD1000304	Seminar	4	

	SWD1000404	Masters Project Phase-2	4	
		Semester Credit:	**20**	
		Total Credits:	**40**	

M.Sc. (Research and Course Work) in Software Development

Sem	Paper Code	Paper Name	Credit	Exit
9	SWD0900204	Deep Learning	4	**M.Sc. (Research and Course Work) in Software Development**
	SWD0900304	Enterprise Resource Planning	4	
	SWD0900404	Internet of Things: Software Development for Embedded Systems	4	
	SWDXXXXX04	Elective-1	4	
	SWD0900504	Masters Project	4	
		Semester Credit:	**20**	
10	SWD1000520	Masters Dissertation	20	
		Semester Credit:	**20**	
		Total Credits:	**40**	

List of Electives:

1. Blockchain Technology
2. Natural Language Processing
3. Image Processing
4. E-Commerce
5. Speech Processing
6. Simulated Robotics
7. No SQL
8. Embedding
9. Any other Course that the DAC decides to offer from time to time based on requirements of contemporary Industries, as well as availability of Faculty Members/Industry Experts